Synthetic Villa

DB-V

Job number	SYN-001
Company profile	iec-default
Design rules	software defaults, not confirmed by the company's engineers

Rev.	Date	Description	Drawn	Checked	Review approved
01	2026-10-31	v1 seed			M2 Verifier
02	2026-10-31				M2 Verifier

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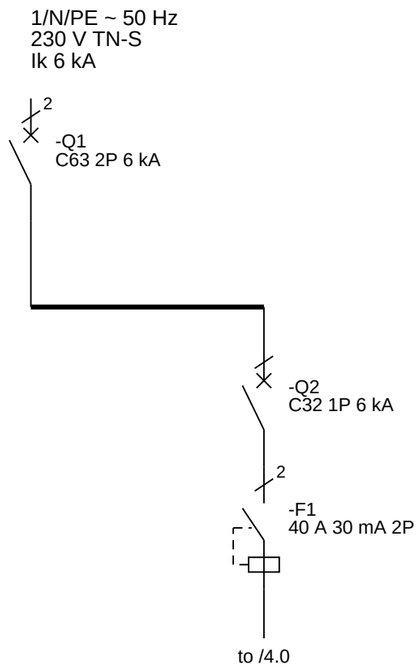

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	Synthetic Villa	SYN-001	DB-V	**Main power**	02

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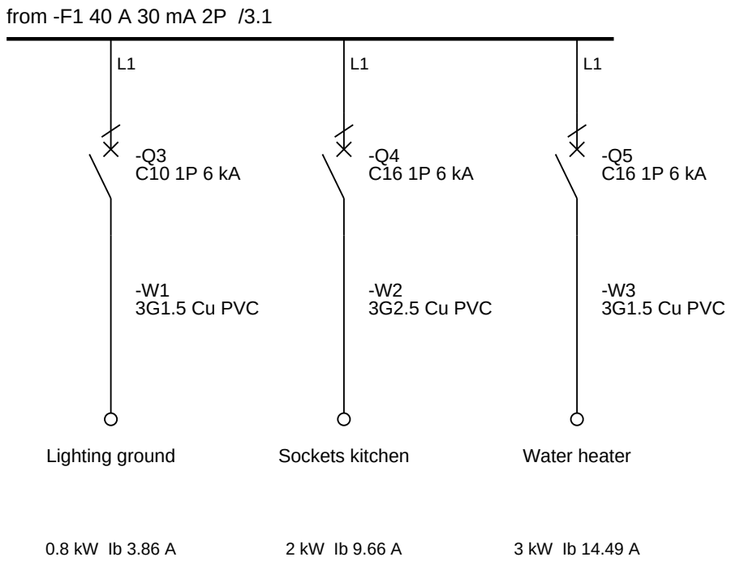

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No.	Note
1	Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.
2	Breakers are rated for at least 6 kA breaking capacity against 6 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too.
3	Leave 1 spare outgoing ways (20 %).
4	Loads without a power factor were taken at cos phi 0.9.
5	Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.
6	1 circuit(s) have no cable length given; their voltage drop is not checked.
7	Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 6 kA from its maker's curve (ABB handbook §2.4).
8	Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.
9	Circuit rules are this software's defaults, not confirmed by the company's engineers.
10	Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 24 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted.

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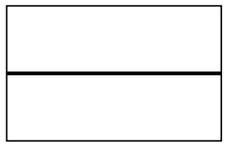
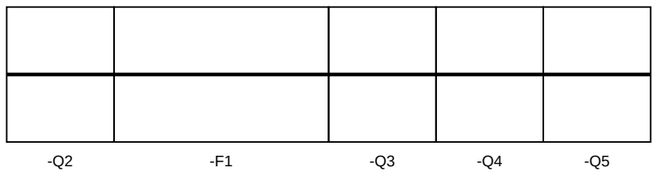
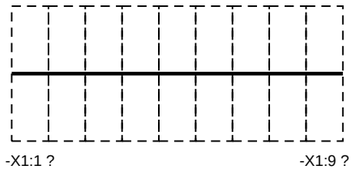
Incomer: 35 mm of rail



Residual current group: 105 mm of rail



Terminals: 0 mm of rail; 9 without a width



Dashed: no width is given for the device's kind (rail_widths_mm in the company settings).

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Terminal	Conductor	Circuit	Cable	mm²	Article
-X1:1	L1	Lighting ground	-W1	1.5	8WH1000-0AF00
-X1:2	N	Lighting ground	-W1	1.5	8WH1000-0AF00
-X1:3	PE	Lighting ground	-W1	1.5	8WH1000-0CF07
-X1:4	L1	Sockets kitchen	-W2	2.5	8WH1000-0AF00
-X1:5	N	Sockets kitchen	-W2	2.5	8WH1000-0AF00
-X1:6	PE	Sockets kitchen	-W2	2.5	8WH1000-0CF07
-X1:7	L1	Water heater	-W3	1.5	8WH1000-0AF00
-X1:8	N	Water heater	-W3	1.5	8WH1000-0AF00
-X1:9	PE	Water heater	-W3	1.5	8WH1000-0CF07

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Cable	From	To	Type	Length
DB-V -W1	DB-V -Q3	Lighting ground	3G1.5 Cu PVC	15 m
DB-V -W2	DB-V -Q4	Sockets kitchen	3G2.5 Cu PVC	12 m
DB-V -W3	DB-V -Q5	Water heater	3G1.5 Cu PVC	

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	Synthetic Villa	SYN-001		**Cable list**	02
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Qty	Description	Article	Order number	Devices
1	Circuit-breaker C63 2P 6 kA	not selected		-Q1
1	Circuit-breaker C32 1P 6 kA	not selected		-Q2
1	Residual current circuit-breaker 40 A 30 mA 2P	not selected		-F1
1	Circuit-breaker C10 1P 6 kA	not selected		-Q3
2	Circuit-breaker C16 1P 6 kA	not selected		-Q4, -Q5
6	Terminal 8WH1	Siemens 8WH1000-0AF00	8WH1000-0AF00	DB-V -X1
3	PE terminal 8WH1	Siemens 8WH1000-0CF07	8WH1000-0CF07	DB-V -X1

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Findings (0 blocking, 26 warning, 6 info), page 1 of 1 — sheet 10 of 10

[WARNING] DB-V: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)
[WARNING] DB-V: 1 circuit(s) have no cable length given; their voltage drop is not checked. (voltage_drop_unchecked; source: policy; engineering confirmation required)
[WARNING] DB-V: Each cable's withstand k²S² is in the cable list; compare it with the selected breaker's let-through I²t at 6 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy; engineering confirmation required)
[WARNING] DB-V: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)
[WARNING] : uses pattern tenant/synth-pe-n-distribution/v1, which no qualified engineer has reviewed; it may not be executed (pattern_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-residual_current_device/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template tenant/synth-dist-bar-3way/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template tenant/synth-dist-bar-3way/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: uses template tenant/synth-strip-pen-2level/v1, which no qualified engineer has reviewed; it may not be executed (template_unreviewed; source: integrity; engineering confirmation required)
[WARNING] DB-V: terminal F of component k-7747a191058bb950aee299ef is in no net and its template does not say whether it must be connected; unresolved (terminal_requiredness_unresolved; source: integrity; engineering confirmation required)
[WARNING] DB-V: terminal F of component k-8788af5120e774e426b7c36a is in no net and its template does not say whether it must be connected; unresolved (terminal_requiredness_unresolved; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-7a62bb919bb2da2ffd501f41's colour is an assumption (a-75adfff7a8a9439e00df4ad6); confirm it (wire_attribute_assumed; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-98ec805f97a0d5d6d6e9b8e1 has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-b37b5548f57074883c15036f has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-b5f4095c51b3ff391915bdfe has no colour; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-7a62bb919bb2da2ffd501f41 has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-98ec805f97a0d5d6d6e9b8e1 has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-b37b5548f57074883c15036f has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-b5f4095c51b3ff391915bdfe has no cross_section_mm2; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-7a62bb919bb2da2ffd501f41 has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[WARNING] DB-V: wire w-b5f4095c51b3ff391915bdfe has no number; no rule sets it yet (wire_attribute_unset; source: integrity; engineering confirmation required)
[INFO] DB-V: Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)
[INFO] DB-V: Breakers are rated for at least 6 kA breaking capacity against 6 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too. (breaking_capacity; source: policy)
[INFO] DB-V: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)
[INFO] DB-V: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)
[INFO] DB-V: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)
[INFO] DB-V: Enclosure: ABB Mini Center compact GDMS332 RX (2CVA250001P0001), 2 rows of 16 modules, H x W x D 498 x 400 x 117 mm, IP41, busbar 200/250 A, 35 kA; 24 modules are left for spare ways (ABB Distribution Boards catalogue (1SKC802027C0201), Mini Center compact, pp. 5, 7). The outgoing terminal strip is not counted. (enclosure_selected; source: policy)